

Ayoub DAOUDI

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AI & ML Engineer — GenAI, RAG, MLOps

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AI & Computer Engineer specialized in LLM/RAG, Machine Learning and MLOps. Experienced in developing end-to-end AI solutions and deploying ML pipelines (API, Docker, AWS). Seeking a work-study position to deepen my production skills.

Education

- **ENSAM Casablanca** Casablanca, Morocco
Engineering Degree – Artificial Intelligence & Computer Engineering *Sep 2019 – Sep 2024*
- **Lycée Boumalne Dades** Tinghir, Morocco
Baccalaureate Diploma – Mathematical Sciences - Merit: Good *Sep 2018 – Jul 2019*

Skills

- **GenAI & LLM:** LangChain, Hugging Face, RAG, Pinecone, FAISS, sentence-transformers, Prompt Engineering.
- **Machine Learning:** Scikit-Learn, XGBoost, PyTorch, TensorFlow, Pandas, NumPy, SMOTE, Optuna, SHAP.
- **MLOps & Cloud:** AWS, Azure, MLFlow, Kubeflow, Docker, Kubernetes, Git, GitHub, GitLab, CI/CD, Jenkins.
- **Languages & Development:** Python, TypeScript, JavaScript, SQL, Flask, Node.js, Next.js, React.
- **Databases:** SQL, NoSQL, PostgreSQL, MongoDB, Graph, Spark, Hadoop, ETL.

Experience

- **Full-Stack & AI Engineer** Casablanca, Morocco
MedCare - Freelance Project *January 2026 – Present*
 - Designed and developed a **RAG** medical assistant (**LangChain**, **Pinecone**, **sentence-transformers** embeddings) integrated into a live healthcare services platform.
 - Developed a **trilingual FR/AR/EN** (full **RTL**) **Next.js 16** application (**React 19**, **TypeScript**) with a multimodal conversational AI agent: **RAG**, **function calling**, **STT/TTS**, emergency detection, symptom triage and automated appointment booking.
 - Deployed the application as **containerized microservices** with **Docker** on **AWS** and set up a **CI/CD** pipeline via **GitHub Actions**.
- **AI & Software Engineer (Internship)** Brest, France
Lab-STICC - IMT Atlantique Brest *Mar 2024 – Aug 2024*
 - Designed and developed a **VS Code extension** (**TypeScript**) detecting the use of **generative AI** tools during coding sessions.
 - Designed detection scenarios and analyzed collected data via **process mining** (**PM4PY**) and variant analysis to compare student groups.
 - Conducted scientific literature review (**Zotero**) and collaborative development under **GitLab** throughout the project.
- **Data Scientist & ML Engineer (Remote Internship)** Indore, India
Technocolabs *Jun 2023 – Oct 2023*
 - Designed and deployed a corporate bankruptcy prediction **ML web application**, from data to production service (**Agile/SCRUM** team, **Jira**).
 - Classification on imbalanced data: **SMOTE** resampling and feature selection (**Mutual Information**, **SelectKBest**, **ExtraTreesClassifier**).
 - Trained and optimized models (**SVM**, **Random Forest**, **XGBoost**) with **GridSearch** and **Optuna**; tracked experiments, metrics and artifacts with **MLflow**.
 - Containerization (**Docker**) and deployment on **AWS EC2** via a **Flask API**; evaluation focused on imbalanced classes (**AUC-ROC**, minority class recall).
- **Data Science Intern** Morocco
Proface *Jul 15 – Sep 15, 2022*
 - First hands-on exposure to **AI/ML** through a 5-day training on core concepts and tools.
 - Contributed to a document recognition system extracting structured data (names, Arabic names, addresses, emails) from text.
 - Manipulated **JSON** metadata for model training using **Python**, **Pandas** and **NumPy**.

Projects

- **Fraud Detection Pipeline** (*in progress*): ML pipeline for fraud detection on imbalanced transactions: resampling (**SMOTE** / under-sampling), feature engineering, model comparison (**Logistic Regression**, **Random Forest**, **XGBoost**), evaluation focused on **AUC-PR** and fraud-class recall. **MLflow** tracking, **SHAP** explainability, containerized API (**FastAPI**, **Docker**), **GitHub Actions CI/CD**.
- **Resummary** (*April 2025*): Full-stack TypeScript/React Router (SSR) resume analysis application: **multimodal LLM** pipeline (**Claude Sonnet 4.6**) vision converting PDF → image via **PDF.js** for OCR-free **ATS** scoring and personalized recommendations, with schema-constrained **prompting** (strict JSON output, repair and timeout fallbacks). **Semantic job matching** built from scratch.**tokenization**, **TF-IDF** vectorization and **cosine similarity** ranking, plus skill extraction with alias normalization to surface matched vs. missing competencies. Authentication, file storage and persistence via **Puter**; job listings retrieval via **RapidAPI JSearch**.
- **MASI Index Predictor** (*Mars 2024*): **Machine learning** forecasting application for the Casablanca Stock Exchange (**MASI**) closing price: **data preprocessing and EDA** with **Pandas**, **Matplotlib** and **Seaborn** (datetime parsing, cleaning, feature selection, trend/seasonality and correlation analysis). Implemented and compared **time-series models**: **ARIMA**, **SARIMA** and **LSTM** against a supervised **linear regression** baseline on **OHLC** features. Model served through a **Flask REST inference API** with **R²**-based confidence scoring.

Languages

- **French**: Fluent **English**: Fluent